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BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation

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https://github.com/salesforce/BLIP

Abstract

Vision-Language Pre-training (VLP) has advanced the performance for many vision-language tasks. However, most existing pre-trained models only excel in either understanding-based tasks or generation-based tasks. Furthermore, performance improvement has been largely achieved by scaling up the dataset with noisy image-text pairs collected from the web, which is a suboptimal source of supervision. In this paper, we propose BLIP, a new VLP framework which transfers flexibly to both vision-language understanding and generation tasks. BLIP effectively utilizes the noisy web data by bootstrapping the captions, where a captioner generates synthetic captions and a filter removes the noisy ones. We achieve state-of-the-art results on a wide range of vision-language tasks, such as image-text retrieval (+2.7% in average recall@1), image captioning (+2.8% in CIDEr), and VQA (+1.6% in VQA score). BLIP also demonstrates strong generalization ability when directly transferred to video-language tasks in a zero-shot manner. Code, models, and datasets are released.

1. Introduction

Vision-language pre-training has recently received tremendous success on various multimodal downstream tasks. However, existing methods have two major limitations:

(1) Model perspective: most methods either adopt an encoder-based model (Radford et al., 2021; Li et al., 2021a), or an encoder-decoder (Cho et al., 2021; Wang et al., 2021) model. However, encoder-based models are less straightforward to directly transfer to text generation tasks (*e.g.* image captioning), whereas encoder-decoder models have not been successfully adopted for image-text retrieval tasks.

(2) Data perspective: most state-of-the-art methods (*e.g.*, CLIP (Radford et al., 2021), ALBEF (Li et al., 2021a), SimVLM (Wang et al., 2021)) pre-train on image-text pairs collected from the web. Despite the performance gain obtained by scaling up the dataset, our paper shows that the noisy web text is suboptimal for vision-language learning.

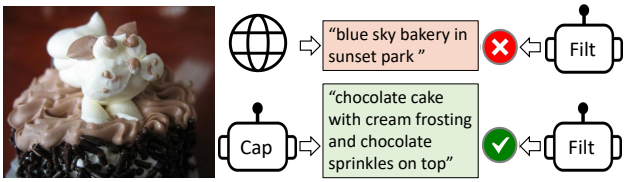


Figure 1. We use a Captioner (Cap) to generate synthetic captions for web images, and a Filter (Filt) to remove noisy captions.

To this end, we propose BLIP: Bootstrapping Language-Image Pre-training for unified vision-language understanding and generation. BLIP is a new VLP framework which enables a wider range of downstream tasks than existing methods. It introduces two contributions from the model and data perspective, respectively:

(a) Multimodal mixture of Encoder-Decoder (MED): a new model architecture for effective multi-task pre-training and flexible transfer learning. An MED can operate either as a unimodal encoder, or an image-grounded text encoder, or an image-grounded text decoder. The model is jointly pre-trained with three vision-language objectives: image-text contrastive learning, image-text matching, and image-conditioned language modeling.

(b) Captioning and Filtering (CapFilt): a new dataset boostrapping method for learning from noisy image-text pairs. We finetune a pre-trained MED into two modules: a *captioner* to produce synthetic captions given web images, and a *filter* to remove noisy captions from both the original web texts and the synthetic texts.

We perform extensive experiments and analysis, and make the following key observations.

- We show that the captioner and the filter work together to achieve substantial performance improvement on various downstream tasks by bootstrapping the captions. We also find that more diverse captions yield larger gains.
- BLIP achieves state-of-the-art performance on a wide range of vision-language tasks, including image-text re-

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BLIP：用于统一视觉-语言理解与生成的自举式语言-图像预训练

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https://github.com/salesforce/BLIP

摘要

视觉-语言预训练（VLP）提升了许多视觉-语言任务的性能。然而，现有的大多数预训练模型只擅长基于理解的任务或基于生成的任务。此外，性能提升很大程度上是通过扩大数据集规模来实现的，这些数据集包含从网络收集的带噪声图像-文本对，而这是一种次优的监督来源。在本文中，我们提出了 BLIP，一种新的 VLP 框架，能够灵活迁移到视觉-语言理解和生成任务。BLIP 通过对字幕进行自举来有效利用带噪声的网络数据，其中字幕生成器生成合成字幕，过滤器则移除有噪声的字幕。我们在广泛的视觉-语言任务上取得了最先进的结果，例如图像-文本检索（平均 recall@1 提升 +2.7%）、图像字幕生成（CIDEr 提升 +2.8%）以及 VQA（VQA 分数提升 +1.6%）。当以零样本方式直接迁移到视频-语言任务时，BLIP 也展现出强大的泛化能力。代码、模型和数据集均已发布。

1. 引言

视觉-语言预训练最近在各种多模态下游任务上取得了巨大成功。然而，现有方法存在两个主要局限性：

（1）模型视角：大多数方法要么采用基于编码器的模型（Radford 等，2021；Li 等，2021a），要么采用编码器-解码器（Cho 等，2021；Wang 等，2021）模型。然而，基于编码器的模型不太容易直接迁移到文本生成任务（*e.g.* 图像描述）中，而编码器-解码器模型尚未成功应用于图文检索任务。

（2）数据视角：大多数最先进的方法（*e.g.*、CLIP（Radford 等，2021）、ALBEF（Li 等，2021a）、SimVLM（Wang 等，2021））在图文对上进行预训练

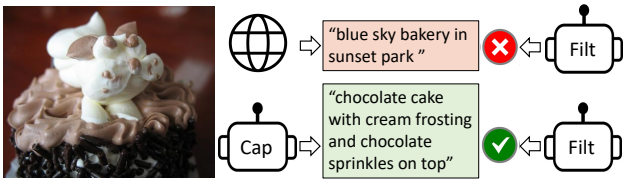


图 1。我们使用描述生成器（Cap）为网络图像生成合成描述，并使用过滤器（Filt）去除有噪声的描述。

收集自网络。尽管通过扩大数据集规模获得了性能提升，但我们的论文表明，嘈杂的网络文本对于视觉-语言学习而言并非最优。

为此，我们提出了 BLIP：用于统一视觉-语言理解与生成的自举式语言-图像预训练。BLIP 是一种新的 VLP 框架，相比现有方法能够支持更广泛的下游任务。它分别从模型和数据角度提出了两项贡献：

(a) 编码器-解码器多模态混合模型（MED）：一种用于有效多任务预训练和灵活迁移学习的新型模型架构。MED 既可以作为单模态编码器运行，也可以作为图像引导的文本编码器运行，或作为图像引导的文本解码器运行。该模型通过三种视觉-语言目标进行联合预训练：图像-文本对比学习、图像-文本匹配以及图像条件语言建模。

(b) 字幕生成与过滤（CapFilt）：一种新的数据集自举方法，用于从有噪声的图文对中学习。我们将预训练的 MED 微调为两个模块：一个 *cap- tioner*，用于根据网络图像生成合成字幕；以及一个 *filter*，用于从原始网络文本和合成文本中去除有噪声的字幕。

我们进行了大量实验和分析，并得出以下关键观察结果。

- 我们表明，字幕生成器和过滤器通过自举字幕协同工作，在各种下游任务上实现了显著的性能提升。我们还发现，字幕越多样化，带来的提升越大。
- BLIP 在广泛的视觉-语言任务上实现了最先进的性能，包括图像-文本 re-

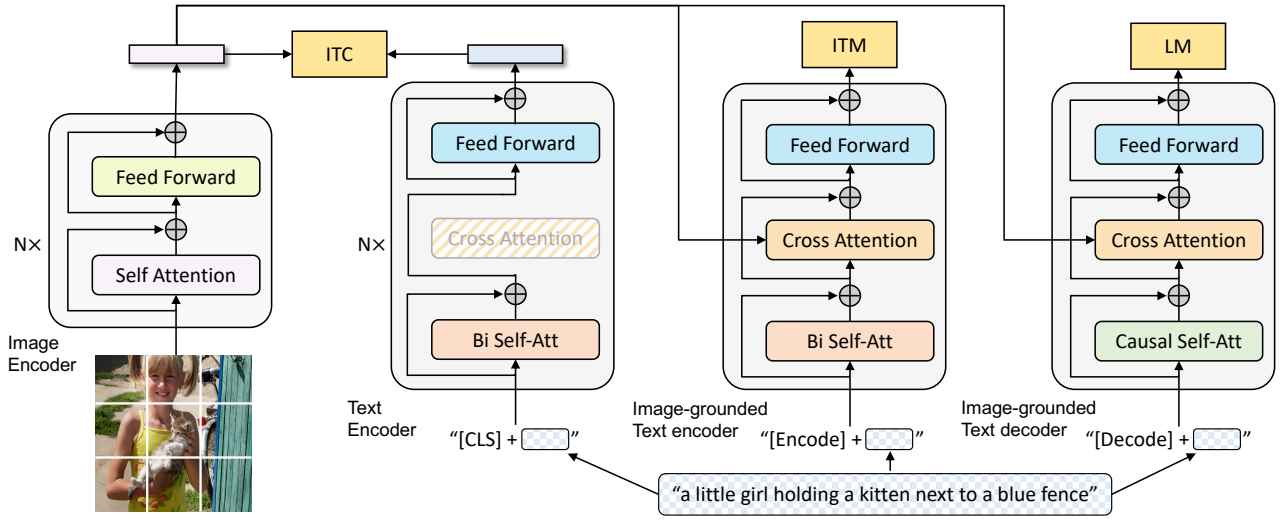


Figure 2. Pre-training model architecture and objectives of BLIP (same parameters have the same color). We propose multimodal mixture of encoder-decoder, a unified vision-language model which can operate in one of the three functionalities: (1) Unimodal encoder is trained with an image-text contrastive (ITC) loss to align the vision and language representations. (2) Image-grounded text encoder uses additional cross-attention layers to model vision-language interactions, and is trained with a image-text matching (ITM) loss to distinguish between positive and negative image-text pairs. (3) Image-grounded text decoder replaces the bi-directional self-attention layers with causal self-attention layers, and shares the same cross-attention layers and feed forward networks as the encoder. The decoder is trained with a language modeling (LM) loss to generate captions given images.

trieval, image captioning, visual question answering, visual reasoning, and visual dialog. We also achieve state-of-the-art zero-shot performance when directly transferring our models to two video-language tasks: text-to-video retrieval and videoQA.

2. Related Work

2.1. Vision-language Pre-training

Vision-language pre-training (VLP) aims to improve performance of downstream vision and language tasks by pre-training the model on large-scale image-text pairs. Due to the prohibitive expense of acquiring human-annotated texts, most methods (Chen et al., 2020; Li et al., 2020; 2021a; Wang et al., 2021; Radford et al., 2021) use image and alt-text pairs crawled from the web (Sharma et al., 2018; Changpinyo et al., 2021; Jia et al., 2021), Despite the use of simple rule-based filters, noise is still prevalent in the web texts. However, the negative impact of the noise has been largely overlooked, shadowed by the performance gain obtained from scaling up the dataset. Our paper shows that the noisy web texts are suboptimal for vision-language learning, and proposes CapFilt that utilizes web datasets in a more effective way.

There have been many attempts to unify various vision and language tasks into a single framework (Zhou et al., 2020; Cho et al., 2021; Wang et al., 2021). The biggest challenge is to design model architectures that can perform both understanding-based tasks (*e.g.* image-text retrieval) and generation-based tasks (*e.g.* image captioning). Neither encoder-based models (Li et al., 2021a;b; Radford et al., 2021) nor encoder-decoder models (Cho et al., 2021; Wang et al., 2021) can excel at both types of tasks, whereas a single unified encoder-decoder (Zhou et al., 2020) also limits the model's capability. Our proposed multimodal mixture of encoder-decoder model offers more flexibility and better performance on a wide range of downstream tasks, in the meantime keeping the pre-training simple and efficient.

2.2. Knowledge Distillation

Knowledge distillation (KD) (Hinton et al., 2015) aims to improve the performance of a student model by distilling knowledge from a teacher model. Self-distillation is a special case of KD where the teacher and student have equal sizes. It has been shown to be effective for image classification (Xie et al., 2020), and recently for VLP (Li et al., 2021a). Different from mostly existing KD methods which simply enforce the student to have the same class predictions as the teacher, our proposed CapFilt can be interpreted as a more effective way to perform KD in the context of VLP, where the captioner distills its knowledge through semantically-rich synthetic captions, and the filter distills its knowledge by removing noisy captions.

2.3. Data Augmentation

While data augmentation (DA) has been widely adopted in computer vision (Shorten & Khoshgoftaar, 2019), DA for language tasks is less straightforward. Recently, generative language models have been used to synthesize examples for various NLP tasks (Kumar et al., 2020; Anaby-Tavor

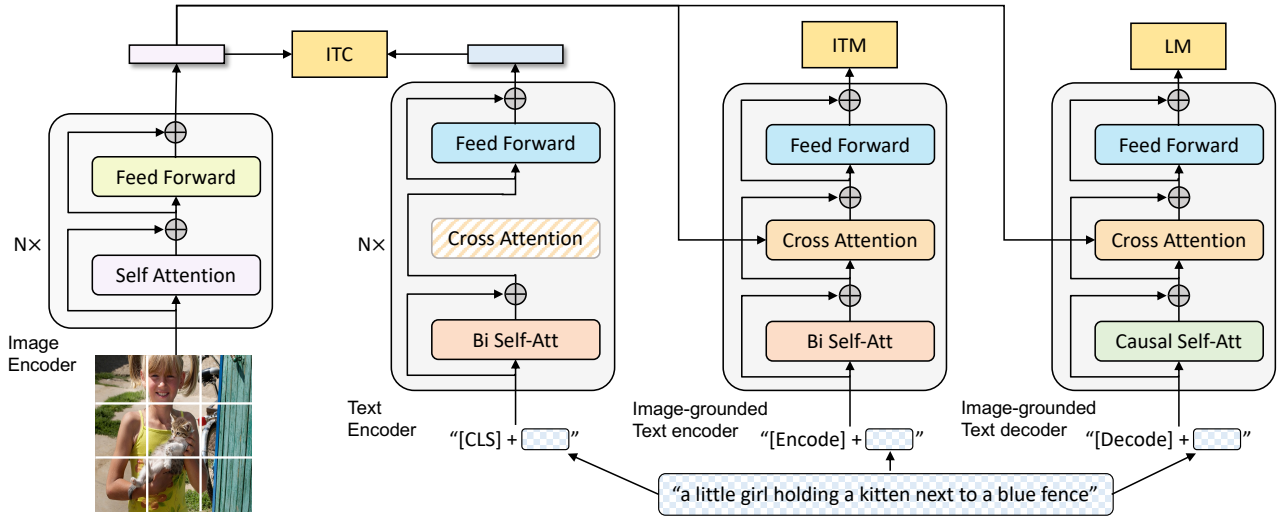


图 2. BLIP 的预训练模型架构与目标（相同参数使用相同颜色表示）。我们提出了多模态编码器-解码器混合架构，这是一种统一的视觉-语言模型，可执行以下三种功能之一：（1）单模态编码器通过图文对比（ITC）损失进行训练，以对齐视觉和语言表征。（2）以图像为基础的文本编码器使用额外的交叉注意力层来建模视觉-语言交互，并通过图文匹配（ITM）损失进行训练，以区分正负图文对。（3）以图像为基础的文本解码器将双向自注意力层替换为因果自注意力层，并与编码器共享相同的交叉注意力层和前馈网络。该解码器通过语言建模（LM）损失进行训练，以根据图像生成描述。

检索、图像描述生成、视觉问答、视觉推理和视觉对话。当将我们的模型直接迁移到两个视频-语言任务（文本到视频检索和视频问答）时，我们也实现了最先进的零样本性能。

2. 相关工作

2.1. 视觉-语言预训练

视觉-语言预训练（VLP）旨在通过在大规模图像-文本对上对模型进行预训练，提升下游视觉和语言任务的性能。由于获取人工标注文本的成本高昂，大多数方法（Chen et al., 2020; Li et al., 2020; 2021a; Wang et al., 2021; Radford et al., 2021）使用从网络爬取的图像及其替代文本（alt-text）对（Sharma et al., 2018; Changpinyo et al., 2021; Jia et al., 2021）。尽管采用了简单的基于规则的过滤器，网络文本中的噪声仍然普遍存在。然而，噪声的负面影响在很大程度上被忽视了，因为它被扩大数据集规模所带来的性能提升所掩盖。我们的论文表明，含噪的网络文本并非视觉-语言学习的最优选择，并提出了 CapFilt，以更有效地利用网络数据集。

已有许多尝试将各种视觉和语言任务统一到一个单一框架中（Zhou et al., 2020；Cho et al., 2021；Wang et al., 2021）。最大的挑战在于设计能够同时执行基于理解的任务（*e.g.* 图文检索）和基于生成的任务（*e.g.* 图像描述生成）的模型架构。两者都不基于编码器的模型（Li et al., 2021a;b; Radford et al., 2021）和编码器-解码器模型（Cho et al., 2021; Wang et al., 2021）都无法在这两类任务上同时表现出色，而单一的统一编码器-解码器（Zhou et al., 2020）也会限制模型的能力。我们提出的多模态编码器-解码器混合模型在广泛的下游任务上提供了更高的灵活性和更好的性能，同时保持预训练的简单高效。

2.2. 知识蒸馏

知识蒸馏（KD）（Hinton et al., 2015）旨在通过从教师模型中蒸馏知识来提升学生模型的性能。自蒸馏是 KD 的一种特殊情况，其中教师模型和学生模型具有相同的规模。它已被证明对图像分类有效（Xie et al., 2020），最近也被证明对 VLP 有效（Li et al., 2021a）。不同于大多数现有的 KD 方法仅仅强制学生模型具有与教师模型相同的类别预测，我们提出的 CapFilt 可以被解释为一种在 VLP 场景下更有效地执行 KD 的方式，其中 captioner 通过语义丰富的合成描述来蒸馏其知识，而 filter 则通过移除有噪声的描述来蒸馏其知识。

2.3. 数据增强

虽然数据增强（DA）已在计算机视觉中被广泛采用（Shorten & Khoshgoftaar, 2019），但用于语言任务的 DA 并不那么直接。最近，生成式语言模型已被用于为各种 NLP 任务合成样本（Kumar et al., 2020；Anaby-Tavor

et al., 2020; Puri et al., 2020; Yang et al., 2020). Different from these methods which focus on the low-resource language-only tasks, our method demonstrates the advantage of synthetic captions in large-scale vision-language pre-training.

3. Method

We propose BLIP, a unified VLP framework to learn from noisy image-text pairs. This section first introduces our new model architecture MED and its pre-training objectives, and then delineates CapFilt for dataset bootstrapping.

3.1. Model Architecture

We employ a visual transformer (Dosovitskiy et al., 2021) as our image encoder, which divides an input image into patches and encodes them as a sequence of embeddings, with an additional `[CLS]` token to represent the global image feature. Compared to using pre-trained object detectors for visual feature extraction (Chen et al., 2020), using a ViT is more computation-friendly and has been adopted by the more recent methods (Li et al., 2021a; Kim et al., 2021).

In order to pre-train a unified model with both understanding and generation capabilities, we propose multimodal mixture of encoder-decoder (MED), a multi-task model which can operate in one of the three functionalities:

(1) **Unimodal encoder**, which separately encodes image and text. The text encoder is the same as BERT (Devlin et al., 2019), where a `[CLS]` token is appended to the beginning of the text input to summarize the sentence.

(2) **Image-grounded text encoder**, which injects visual information by inserting one additional cross-attention (CA) layer between the self-attention (SA) layer and the feed forward network (FFN) for each transformer block of the text encoder. A task-specific `[Encode]` token is appended to the text, and the output embedding of `[Encode]` is used as the multimodal representation of the image-text pair.

(3) **Image-grounded text decoder**, which replaces the bi-directional self-attention layers in the image-grounded text encoder with causal self-attention layers. A `[Decode]` token is used to signal the beginning of a sequence, and an end-of-sequence token is used to signal its end.

3.2. Pre-training Objectives

We jointly optimize three objectives during pre-training, with two understanding-based objectives and one generation-based objective. Each image-text pair only requires one forward pass through the computational-heavier visual transformer, and three forward passes through the text transformer, where different functionalities are activated to compute the three losses as delineated below.

Image-Text Contrastive Loss (ITC) activates the unimodal encoder. It aims to align the feature space of the visual transformer and the text transformer by encouraging positive image-text pairs to have similar representations in contrast to the negative pairs. It has been shown to be an effective objective for improving vision and language understanding (Radford et al., 2021; Li et al., 2021a). We follow the ITC loss by Li et al. (2021a), where a momentum encoder is introduced to produce features, and soft labels are created from the momentum encoder as training targets to account for the potential positives in the negative pairs.

Image-Text Matching Loss (ITM) activates the image-grounded text encoder. It aims to learn image-text multimodal representation that captures the fine-grained alignment between vision and language. ITM is a binary classification task, where the model uses an ITM head (a linear layer) to predict whether an image-text pair is positive (matched) or negative (unmatched) given their multimodal feature. In order to find more informative negatives, we adopt the hard negative mining strategy by Li et al. (2021a), where negatives pairs with higher contrastive similarity in a batch are more likely to be selected to compute the loss.

Language Modeling Loss (LM) activates the image-grounded text decoder, which aims to generate textual descriptions given an image. It optimizes a cross entropy loss which trains the model to maximize the likelihood of the text in an autoregressive manner. We apply a label smoothing of 0.1 when computing the loss. Compared to the MLM loss that has been widely-used for VLP, LM enables the model with the generalization capability to convert visual information into coherent captions.

In order to perform efficient pre-training while leveraging multi-task learning, the text encoder and text decoder share all parameters except for the SA layers. The reason is that the differences between the encoding and decoding tasks are best captured by the SA layers. In particular, the encoder employs *bi-directional* self-attention to build representations for the *current* input tokens, while the decoder employs *causal* self-attention to predict *next* tokens. On the other hand, the embedding layers, CA layers and FFN function similarly between encoding and decoding tasks, therefore sharing these layers can improve training efficiency while benefiting from multi-task learning,

3.3. CapFilt

Due to the prohibitive annotation cost, there exist a limited number of high-quality human-annotated image-text pairs $\{(I_h, T_h)\}$ (e.g., COCO (Lin et al., 2014)). Recent work (Li et al., 2021a; Wang et al., 2021) utilizes a much larger number of image and alt-text pairs $\{(I_w, T_w)\}$ that are automatically collected from the web. However, the alt-texts often do not accurately describe the visual content of the images, making them a noisy signal that is suboptimal for learning vision-language alignment.

等，2020；Puri 等，2020；Yang 等，2020）。与这些专注于仅低资源语言任务的方法不同，我们的方法展示了合成字幕在大规模视觉-语言预训练中的优势。

3. 方法

我们提出了 BLIP，一个用于从噪声图文对中学习的统一 VLP 框架。本节首先介绍我们的新模型架构 MED 及其预训练目标，然后阐述用于数据集自举的 CapFilt。

3.1. 模型架构

我们采用视觉 Transformer（Dosovitskiy et al., 2021）作为图像编码器，它将输入图像划分为图像块，并将其编码为一系列嵌入，同时添加一个 [CLS] 标记来表示全局图像特征。与使用预训练目标检测器进行视觉特征提取相比（Chen et al., 2020），使用 ViT 在计算上更加友好，并已被较新的方法所采用（Li et al., 2021a; Kim et al., 2021）。

为了预训练一个同时具备理解和生成能力的统一模型，我们提出了多模态编码器-解码器混合模型（MED），这是一个多任务模型，可在以下三种功能之一中运行：

(1) 单模态编码器，分别对图像和文本进行编码。文本编码器与 BERT（Devlin 等，2019）相同，其中在文本输入的开头添加一个 [CLS] 标记，用于概括句子。

(2) 图像锚定文本编码器，通过在文本编码器每个 Transformer 块的自注意力（SA）层和前馈网络（FFN）之间插入一个额外的交叉注意力（CA）层来注入视觉信息。一个任务特定的 [Encode] 标记被追加到文本中，[Encode] 的输出嵌入被用作图像-文本对的多模态表示。

（3）基于图像的文本解码器，它将基于图像的文本编码器中的双向自注意力层替换为因果自注意力层。使用 [Decode] 标记表示序列的开始，并使用序列结束标记表示其结束。

3.2. 预训练目标

在预训练期间，我们联合优化三个目标，其中两个是基于理解的目标，一个是基于生成的目标。每个图文对只需通过计算开销较大的视觉 Transformer 进行一次前向传播，并通过文本 Transformer 进行三次前向传播，在此过程中会激活不同的功能以计算如下所述的三个损失。

图像-文本对比损失（ITC）激活单模态编码器。它旨在对齐视觉转换-的特征空间。通过鼓励正向图文对具有相似的表征、而与负向图文对形成对比，来训练前者和文本 Transformer。事实证明，这是一种提升视觉与语言理解能力的有效目标（Radford et al., 2021; Li et al., 2021a）。我们遵循 Li et al.（2021a）的 ITC 损失，其中引入了动量编码器来生成特征，并由动量编码器创建软标签作为训练目标，以考虑负样本对中潜在的正样本。

图文匹配损失（ITM）激活以图像为基础的文本编码器。其目标是学习能够捕捉视觉与语言之间细粒度对齐关系的图文多模态表示。ITM 是一项二分类任务，其中模型使用 ITM 头（一个线性层）根据图文对的多模态特征来预测该图文对是正样本（匹配）还是负样本（不匹配）。为了找到信息量更大的负样本，我们采用了 Li 等人（2021a）的困难负样本挖掘策略，即在一个批次中，对比相似度更高的负样本对更有可能被选中用于计算损失。

语言建模损失（LM）激活了图像驱动的文本解码器，其目标是在给定图像的情况下生成文本描述。它优化交叉熵损失，以自回归的方式训练模型最大化文本的似然。在计算损失时，我们应用 0.1 的标签平滑。与已广泛用于 VLP 的 MLM 损失相比，LM 使模型具备将视觉信息转换为连贯描述的泛化能力。

为了在利用多任务学习的同时进行高效的预训练，文本编码器和文本解码器共享除 SA 层之外的所有参数。原因在于，编码任务与解码任务之间的差异最适合由 SA 层来捕捉。具体而言，编码器采用 *bi-directional* 自注意力为 *current* 输入词元构建表示，而解码器采用 *causal* 自注意力来预测 *next* 词元。另一方面，嵌入层、CA 层和 FFN 在编码任务与解码任务中的功能相似，因此共享这些层既能提高训练效率，又能受益于多任务学习。

3.3. CapFilt

由于标注成本高昂，高质量的人工标注图文对数量有限$\{(I_h, T_h)\}$ (，例如 COCO（Lin 等，2014)）。近期工作（Li 等，2021a；Wang 等，2021）利用了数量大得多的图像与替代文本对$\{(I_w, T_w)\}$，这些数据是从网络上自动收集的。然而，替代文本往往无法准确描述图像的视觉内容，使其成为一种有噪声的信号，不利于学习视觉-语言对齐。

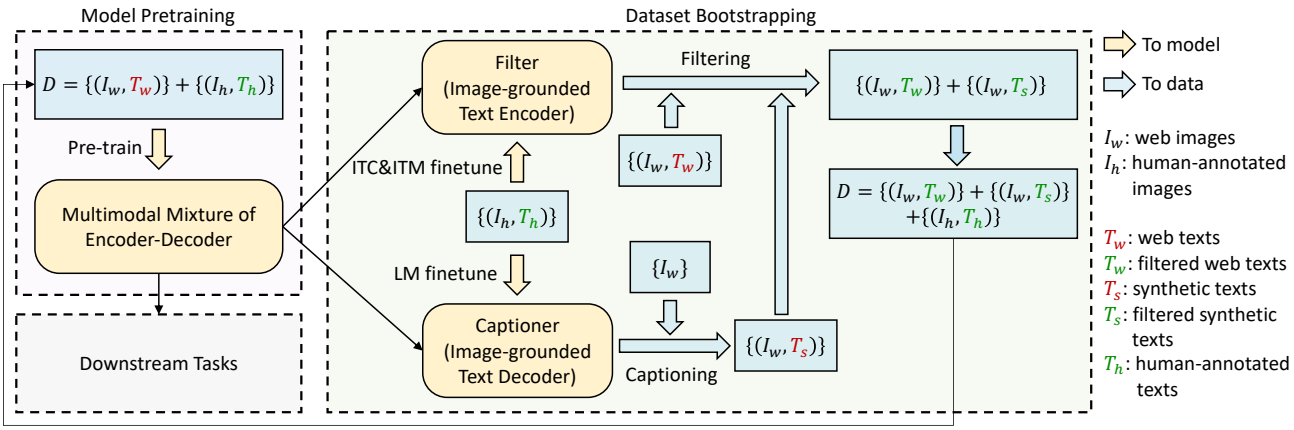


Figure 3. Learning framework of BLIP. We introduce a captioner to produce synthetic captions for web images, and a filter to remove noisy image-text pairs. The captioner and filter are initialized from the same pre-trained model and finetuned individually on a small-scale human-annotated dataset. The bootstrapped dataset is used to pre-train a new model.

We propose Captioning and Filtering (CapFilt), a new method to improve the quality of the text corpus. Figure 3 gives an illustration of CapFilt. It introduces two modules: a *captioner* to generate captions given web images, and a *filter* to remove noisy image-text pairs. Both the captioner and the filter are initialized from the same pre-trained MED model, and finetuned individually on the COCO dataset. The finetuning is a lightweight procedure.

Specifically, the *captioner* is an image-grounded text decoder. It is finetuned with the LM objective to decode texts given images. Given the web images I_w, the captioner generates synthetic captions T_s with one caption per image. The *filter* is an image-grounded text encoder. It is finetuned with the ITC and ITM objectives to learn whether a text matches an image. The filter removes noisy texts in both the original web texts T_w and the synthetic texts T_s, where a text is considered to be noisy if the ITM head predicts it as unmatched to the image. Finally, we combine the filtered image-text pairs with the human-annotated pairs to form a new dataset, which we use to pre-train a new model.

4. Experiments and Discussions

In this section, we first introduce pre-training details. Then we provide a detailed experimental analysis on our method.

4.1. Pre-training Details

Our models are implemented in PyTorch (Paszke et al., 2019) and pre-trained on two 16-GPU nodes. The image transformer is initialized from ViT pre-trained on ImageNet (Touvron et al., 2020; Dosovitskiy et al., 2021), and the text transformer is initialized from $\text{BERT}_{\text{base}}$ (Devlin et al., 2019). We explore two variants of ViTs: ViT-B/16 and ViT-L/16. Unless otherwise specified, all results reported in this paper as "BLIP" uses ViT-B. We pre-train the model for 20 epochs using a batch size of 2880 (ViT-B) / 2400 (ViT-L). We use AdamW (Loshchilov & Hutter, 2017) optimizer with a weight decay of 0.05. The learning rate is warmed-up to $3e$-4 (ViT-B) / $2e$-4 (ViT-L) and decayed linearly with a rate of 0.85. We take random image crops of resolution 224×224 during pre-training, and increase the image resolution to 384×384 during finetuning. We use the same pre-training dataset as Li et al. (2021a) with 14M images in total, including two human-annotated datasets (COCO and Visual Genome (Krishna et al., 2017)), and three web datasets (Conceptual Captions (Changpinyo et al., 2021), Conceptual 12M (Changpinyo et al., 2021), SBU captions (Ordonez et al., 2011)). We also experimented with an additional web dataset, LAION (Schuhmann et al., 2021), which contains 115M images with more noisy texts[1]. More details about the datasets can be found in the appendix.

4.2. Effect of CapFilt

In Table 1, we compare models pre-trained on different datasets to demonstrate the efficacy of CapFilt on downstream tasks, including image-text retrieval and image captioning with finetuned and zero-shot settings.

When only the captioner or the filter is applied to the dataset with 14M images, performance improvement can be observed. When applied together, their effects compliment each other, leading to substantial improvements compared to using the original noisy web texts.

CapFilt can further boost performance with a larger dataset and a larger vision backbone, which verifies its scalability in both the data size and the model size. Furthermore, by using a large captioner and filter with ViT-L, performance of the base model can also be improved.

[1]We only download images whose shorter edge is larger than 256 pixels from the original LAION400M. Due to the large size of LAION, we only use $1/5$ of it each epoch during pre-training.

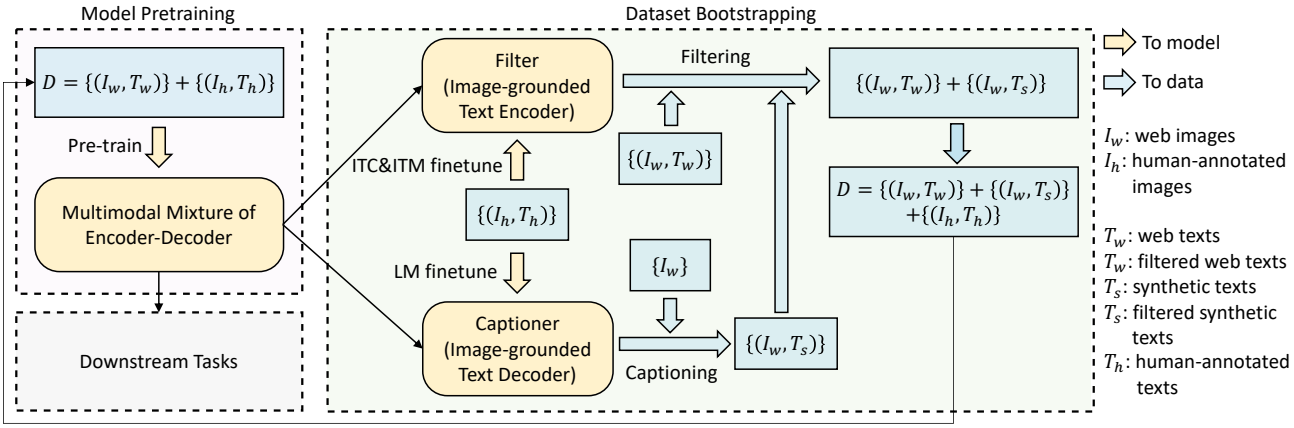


图 3. BLIP 的学习框架。我们引入一个字幕生成器来为网络图像生成合成字幕，并引入一个过滤器来移除有噪声的图像-文本对。字幕生成器和过滤器由同一个预训练模型初始化，并分别在小规模人工标注数据集上进行微调。自举数据集用于预训练一个新模型。

我们提出了 Captioning and Filtering（CapFilt），一种用于提升文本语料库质量的新方法。图 3 展示了 CapFilt 的示意图。它引入了两个模块：一个 *captioner*，用于根据网络图像生成描述；以及一个 *filter*，用于移除有噪声的图像-文本对。描述生成器和过滤器均由同一个预训练的 MED 模型初始化，并分别在 COCO 数据集上进行微调。微调是一个轻量级过程。

具体而言，*captioner* 是一个基于图像的文本解码器。它使用 LM 目标进行微调，以在给定图像的情况下解码文本。给定网络图像 I_w，字幕生成器会生成合成字幕 T_s，每张图像对应一条字幕。*filter* 是一个基于图像的文本编码器。它使用 ITC 和 ITM 目标进行微调，以学习判断文本是否与图像匹配。过滤器会移除原始网络文本 T_w 和合成文本 T_s 中的噪声文本；如果 ITM 头预测某段文本与图像不匹配，则该文本被视为噪声文本。最后，我们将过滤后的图像-文本对与人工标注的文本对结合起来，形成一个新的数据集，并用它来预训练一个新模型。

4. 实验与讨论

在本节中，我们首先介绍预训练细节。然后，我们对我们的方法进行了详细的实验分析。

4.1. 预训练细节

我们的模型在 PyTorch（Paszke 等，2019）中实现，并在两个 16-GPU 节点上进行预训练。图像 Transformer 使用在 ImageNet 上预训练的 ViT 进行初始化（Touvron 等，2020；Dosovitskiy 等，2021），文本 Transformer 使用 $\text{BERT}_{\text{base}}$ (Devlin 等，2019) 进行初始化。我们探索了 ViT 的两个变体：ViT-B/16 和 ViT-L/16。除非另有说明，本文中报告为“BLIP”的所有结果均使用 ViT-B。我们使用 2880（ViT-B）/ 的批量大小对模型进行 20 个 epoch 的预训练。

2400（ViT-L）。我们使用 AdamW（Loshchilov & Hutter，2017）优化器，权重衰减为 0.05。学习率预热至 3e-4（ViT-B）/ 2e-4（ViT-L），并以 0.85 的速率线性衰减。在预训练期间，我们采用分辨率为 224 × 224 的随机图像裁剪，并在微调期间将图像分辨率提高到 384 × 384。我们使用与 Li 等人（2021a）相同的预训练数据集，总共包含 1400 万张图像，包括两个由人工标注的数据集（COCO 和 Visual Genome（Krishna 等人，2017）），以及三个网络数据集（Conceptual Captions（Changpinyo 等人，2021）、Conceptual 12M（Changpinyo 等人，2021）、SBU captions（Ordonez 等人，2011））。我们还实验了一个额外的网络数据集 LAION（Schuhmann 等人，2021），其中包含 1.15 亿张图像，并带有更多噪声文本[1]。有关数据集的更多细节可见附录。

4.2. CapFilt 的影响

在表 1 中，我们比较了在不同数据集上预训练的模型，以展示 CapFilt 在下游任务中的有效性，包括微调和零样本设置下的图文检索与图像描述生成。

当仅将描述生成器或过滤器应用于包含 1400 万张图像的数据集时，可以观察到性能提升。当二者同时应用时，它们的效果相互补充，与使用原始嘈杂的网络文本相比，带来了显著提升。

CapFilt 可以通过更大的数据集和更大的视觉骨干网络进一步提升性能，这验证了其在数据规模和模型规模上的可扩展性。此外，通过使用大型描述生成器以及带有 ViT-L 的过滤器，基础模型的性能也可以得到提升。

[1]We only download images whose shorter edge is larger than 256 pixels from the original LAION400M. Due to the large size of LAION, we only use $1/5$ of it each epoch during pre-training.

Pre-train dataset	Bootstrap C	Bootstrap F	Vision backbone	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
COCO+VG +CC+SBU (14M imgs)	✗	✗	ViT-B/16	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
	✗	✓$_B$		79.1	61.5	94.1	82.8	38.1	128.2	102.7	14.0
	✓$_B$	✗		79.7	62.0	94.4	83.6	38.4	128.9	103.4	14.2
	✓$_B$	✓$_B$		80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4
COCO+VG +CC+SBU +LAION (129M imgs)	✗	✗	ViT-B/16	79.6	62.0	94.3	83.6	38.8	130.1	105.4	14.2
	✓$_B$	✓$_B$		81.9	64.3	96.0	85.0	39.4	131.4	106.3	14.3
	✓$_L$	✓$_L$		81.2	64.1	96.0	85.5	39.7	133.3	109.6	14.7
	✗	✗	ViT-L/16	80.6	64.1	95.1	85.5	40.3	135.5	112.5	14.7
	✓$_L$	✓$_L$		82.4	65.1	96.7	86.7	40.4	136.7	113.2	14.8

Table 1. Evaluation of the effect of the captioner (C) and filter (F) for dataset bootstrapping. Downstream tasks include image-text retrieval and image captioning with finetuning (FT) and zero-shot (ZS) settings. TR / IR@1: recall@1 for text retrieval / image retrieval. ✓$_{B/L}$: captioner or filter uses ViT-B / ViT-L as vision backbone.

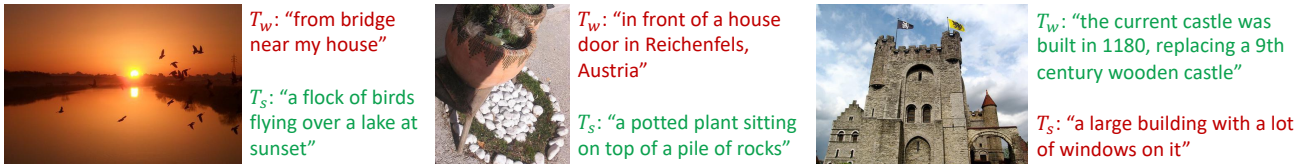


Figure 4. Examples of the web text T_w and the synthetic text T_s. Green texts are accepted by the filter, whereas red texts are rejected.

Generation method	Noise ratio	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
None	N.A.	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
Beam	19%	79.6	61.9	94.1	83.1	38.4	128.9	103.5	14.2
Nucleus	25%	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

Table 2. Comparison between beam search and nucleus sampling for synthetic caption generation. Models are pre-trained on 14M images.

Layers shared	#parameters	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
All	224M	77.3	59.5	93.1	81.0	37.2	125.9	100.9	13.1
All except CA	252M	77.5	59.9	93.1	81.3	37.4	126.1	101.2	13.1
All except SA	252M	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
None	361M	78.3	60.5	93.6	81.9	37.8	127.4	101.8	13.9

Table 3. Comparison between different parameter sharing strategies for the text encoder and decoder during pre-training.

In Figure 4, we show some example captions and their corresponding images, which qualitatively demonstrate the effect of the captioner to generate new textual descriptions, and the filter to remove noisy captions from both the original web texts and the synthetic texts. More examples can be found in the appendix.

4.3. Diversity is Key for Synthetic Captions

In CapFilt, we employ nucleus sampling (Holtzman et al., 2020) to generate synthetic captions. Nucleus sampling is a stochastic decoding method, where each token is sampled from a set of tokens whose cumulative probability mass exceeds a threshold p ($p = 0.9$ in our experiments). In Table 2, we compare it with beam search, a deterministic decoding method which aims to generate captions with the highest probability. Nucleus sampling leads to evidently better performance, despite being more noisy as suggested by a higher noise ratio from the filter. We hypothesis that the reason is that nucleus sampling generates more diverse and surprising captions, which contain more new information that the model could benefit from. On the other hand, beam search tends to generate safe captions that are common in the dataset, hence offering less extra knowledge.

4.4. Parameter Sharing and Decoupling

During pre-training, the text encoder and decoder share all parameters except for the self-attention layers. In Table 3, we evaluate models pre-trained with different parameter sharing strategies, where pre-training is performed on the 14M images with web texts. As the result shows, sharing all

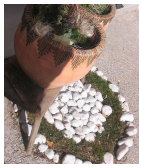
Pre-train dataset	Bootstrap C	Bootstrap F	Vision backbone	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
COCO+VG +CC+SBU (14M imgs)	✗	✗	ViT-B/16	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
	✗	✓$_B$		79.1	61.5	94.1	82.8	38.1	128.2	102.7	14.0
	✓$_B$	✗		79.7	62.0	94.4	83.6	38.4	128.9	103.4	14.2
	✓$_B$	✓$_B$		80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4
COCO+VG +CC+SBU +LAION (129M imgs)	✗	✗	ViT-B/16	79.6	62.0	94.3	83.6	38.8	130.1	105.4	14.2
	✓$_B$	✓$_B$		81.9	64.3	96.0	85.0	39.4	131.4	106.3	14.3
	✓$_L$	✓$_L$		81.2	64.1	96.0	85.5	39.7	133.3	109.6	14.7
	✗	✗	ViT-L/16	80.6	64.1	95.1	85.5	40.3	135.5	112.5	14.7
	✓$_L$	✓$_L$		82.4	65.1	96.7	86.7	40.4	136.7	113.2	14.8

表 1. 评估字幕生成器（C）和过滤器（F）对数据集自举的影响。下游任务包括在微调（FT）和零样本（ZS）设置下的图文检索和图像描述生成。TR / IR@1：文本检索 / 图像检索的 recall@1。✓$_{B/L}$：字幕生成器或过滤器使用 ViT-B / ViT-L 作为视觉骨干网络。



T_w: "from bridge near my house"

T_s: "a flock of birds flying over a lake at sunset"



T_w："在一所房子前 door 在赖兴费尔斯，Austria"

T_s："一盆盆栽植物摆放着o在一堆岩石上"



T_w："现在的城堡建于1180年，取代了一座9世纪的木制城堡" T_s："一栋有很多窗户的大型建筑"

图 4. 网络文本 T_w 和合成文本 T_s 的示例。绿色文本被过滤器接受，而红色文本被拒绝。

Generation method	Noise ratio	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
None	N.A.	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
Beam	19%	79.6	61.9	94.1	83.1	38.4	128.9	103.5	14.2
Nucleus	25%	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

表2. 用于合成标题生成的束搜索与核采样的比较。模型在1400万张图像上进行了预训练。

Layers shared	#parameters	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
All	224M	77.3	59.5	93.1	81.0	37.2	125.9	100.9	13.1
All except CA	252M	77.5	59.9	93.1	81.3	37.4	126.1	101.2	13.1
All except SA	252M	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
None	361M	78.3	60.5	93.6	81.9	37.8	127.4	101.8	13.9

表 3. 预训练期间文本编码器和解码器不同参数共享策略的比较。

在图 4 中，我们展示了一些示例字幕及其对应的图像，定性地说明了字幕生成器生成新的文本描述的效果，以及过滤器从原始网络文本和合成文本中去除噪声字幕的效果。更多示例见附录。

4.3. 多样性是合成字幕的关键

在 CapFilt 中，我们采用核采样（Holtzman 等，2020）来生成合成字幕。核采样是一种随机解码方法，其中每个词元都从一组词元中采样，这些词元的累积概率质量超过阈值 p ($p = 0.9$（在我们的实验中））。在表 2 中，我们将其与束搜索进行比较，束搜索是一种确定性解码方法，旨在生成具有以下特征的字幕最高概率。尽管根据过滤器给出的更高噪声比例来看，核采样的噪声更大，但它带来了明显更好的性能。我们假设其原因在于，核采样生成的描述更加多样且更具惊喜性，其中包含更多模型可以从中受益的新信息。另一方面，束搜索往往会生成数据集中常见的安全描述，因此提供的额外知识较少。

4.4. 参数共享与解耦

在预训练期间，文本编码器和解码器共享除自注意力层之外的所有参数。在表 3 中，我们评估了采用不同参数共享策略进行预训练的模型，其中预训练是在带有网络文本的 1400 万张图像上进行的。结果显示，共享所有

Captioner & Filter	Noise ratio	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
Share parameters	8%	79.8	62.2	94.3	83.7	38.4	129.0	103.5	14.2
Decoupled	25%	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

Table 4. Effect of sharing parameters between the captioner and filter. Models are pre-trained on 14M images.

Method	Pre-train # Images	COCO (5K test set) TR R@1	TR R@5	TR R@10	IR R@1	IR R@5	IR R@10	Flickr30K (1K test set) TR R@1	TR R@5	TR R@10	IR R@1	IR R@5	IR R@10
UNITER (Chen et al., 2020)	4M	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA (Gan et al., 2020)	4M	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
OSCAR (Li et al., 2020)	4M	70.0	91.1	95.5	54.0	80.8	88.5	-	-	-	-	-	-
UNIMO (Li et al., 2021b)	5.7M	-	-	-	-	-	-	89.4	98.9	99.8	78.0	94.2	97.1
ALIGN (Jia et al., 2021)	1.8B	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	100.0	84.9	97.4	98.6
ALBEF (Li et al., 2021a)	14M	77.6	94.3	97.2	60.7	84.3	90.5	95.9	99.8	100.0	85.6	97.5	98.9
BLIP	14M	80.6	95.2	97.6	63.1	85.3	91.1	96.6	99.8	**100.0**	87.2	97.5	98.8
BLIP	129M	**81.9**	95.4	97.8	**64.3**	85.7	91.5	**97.3**	**99.9**	**100.0**	87.3	97.6	**98.9**
$\text{BLIP}_{\text{CapFilt-L}}$	129M	81.2	**95.7**	**97.9**	64.1	**85.8**	**91.6**	97.2	**99.9**	**100.0**	**87.5**	**97.7**	**98.9**
$\text{BLIP}_{\text{ViT-L}}$	129M	82.4	95.4	97.9	65.1	86.3	91.8	97.4	99.8	99.9	87.6	97.7	99.0

Table 5. Comparison with state-of-the-art image-text retrieval methods, finetuned on COCO and Flickr30K datasets. $\text{BLIP}_{\text{CapFilt-L}}$ pre-trains a model with ViT-B backbone using a dataset bootstrapped by captioner and filter with ViT-L.

Method	Pre-train # Images	Flickr30K (1K test set) TR R@1	TR R@5	TR R@10	IR R@1	IR R@5	IR R@10
CLIP	400M	88.0	98.7	99.4	68.7	90.6	95.2
ALIGN	1.8B	88.6	98.7	99.7	75.7	93.8	96.8
ALBEF	14M	94.1	99.5	99.7	82.8	96.3	98.1
BLIP	14M	94.8	99.7	**100.0**	84.9	96.7	98.3
BLIP	129M	**96.0**	**99.9**	**100.0**	85.0	**96.8**	98.6
$\text{BLIP}_{\text{CapFilt-L}}$	129M	**96.0**	**99.9**	**100.0**	**85.5**	**96.8**	**98.7**
$\text{BLIP}_{\text{ViT-L}}$	129M	96.7	100.0	100.0	86.7	97.3	98.7

Table 6. Zero-shot image-text retrieval results on Flickr30K.

layers except for SA leads to better performance compared to not sharing, while also reducing the model size thus improveing training efficiency. If the SA layers are shared, the model's performance would degrade due to the conflict between the encoding task and the decoding task.

During CapFilt, the captioner and the filter are end-to-end finetuned individually on COCO. In Table 4, we study the effect if the captioner and filter share parameters in the same way as pre-training. The performance on the downstream tasks decreases, which we mainly attribute to *confirmation bias*. Due to parameter sharing, noisy captions produced by the captioner are less likely to be filtered out by the filter, as indicated by the lower noise ratio (8% compared to 25%).

5. Comparison with State-of-the-arts

In this section, we compare BLIP to existing VLP methods on a wide range of vision-language downstream tasks[2]. Next we briefly introduce each task and finetuning strategy. More details can be found in the appendix.

5.1. Image-Text Retrieval

We evaluate BLIP for both image-to-text retrieval (TR) and text-to-image retrieval (IR) on COCO and Flickr30K (Plummer et al., 2015) datasets. We finetune the pre-trained model using ITC and ITM losses. To enable faster inference speed, we follow Li et al. (2021a) and first select k candidates based on the image-text feature similarity, and then rerank the selected candidates based on their pairwise ITM scores. We set $k = 256$ for COCO and $k = 128$ for Flickr30K.

As shown in Table 5, BLIP achieves substantial performance improvement compared with existing methods. Using the same 14M pre-training images, BLIP outperforms the previous best model ALBEF by +2.7% in average recall@1 on COCO. We also perform zero-shot retrieval by directly transferring the model finetuned on COCO to Flickr30K. The result is shown in Table 6, where BLIP also outperforms existing methods by a large margin.

5.2. Image Captioning

We consider two datasets for image captioning: NoCaps (Agrawal et al., 2019) and COCO, both evaluated using the model finetuned on COCO with the LM loss. Similar as Wang et al. (2021), we add a prompt "a picture of" at the beginning of each caption, which leads to slightly better results. As shown in Table 7, BLIP with 14M pre-training images substantially outperforms methods using a similar amount of pre-training data. BLIP with 129M images achieves competitive performance as LEMON with

[2] we omit SNLI-VE from the benchmark because its test data has been reported to be noisy (Do et al., 2020)

Captioner & Filter	Noise ratio	Retrieval-FT (COCO) TR@1	Retrieval-FT (COCO) IR@1	Retrieval-ZS (Flickr) TR@1	Retrieval-ZS (Flickr) IR@1	Caption-FT (COCO) B@4	Caption-FT (COCO) CIDEr	Caption-ZS (NoCaps) CIDEr	Caption-ZS (NoCaps) SPICE
Share parameters	8%	79.8	62.2	94.3	83.7	38.4	129.0	103.5	14.2
Decoupled	25%	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

表 4. 在字幕生成器和过滤器之间共享参数的效果。模型在 1400 万张图像上进行了预训练。

Method	Pre-train # Images	COCO (5K test set) TR R@1	TR R@5	TR R@10	IR R@1	IR R@5	IR R@10	Flickr30K (1K test set) TR R@1	TR R@5	TR R@10	IR R@1	IR R@5	IR R@10
UNITER (Chen et al., 2020)	4M	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA (Gan et al., 2020)	4M	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
OSCAR (Li et al., 2020)	4M	70.0	91.1	95.5	54.0	80.8	88.5	-	-	-	-	-	-
UNIMO (Li et al., 2021b)	5.7M	-	-	-	-	-	-	89.4	98.9	99.8	78.0	94.2	97.1
ALIGN (Jia et al., 2021)	1.8B	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	100.0	84.9	97.4	98.6
ALBEF (Li et al., 2021a)	14M	77.6	94.3	97.2	60.7	84.3	90.5	95.9	99.8	100.0	85.6	97.5	98.9
BLIP	14M	80.6	95.2	97.6	63.1	85.3	91.1	96.6	99.8	**100.0**	87.2	97.5	98.8
BLIP	129M	**81.9**	95.4	97.8	**64.3**	85.7	91.5	**97.3**	**99.9**	**100.0**	87.3	97.6	**98.9**
$\text{BLIP}_{\text{CapFilt-L}}$	129M	81.2	**95.7**	**97.9**	64.1	**85.8**	**91.6**	97.2	**99.9**	**100.0**	**87.5**	**97.7**	**98.9**
$\text{BLIP}_{\text{ViT-L}}$	129M	82.4	95.4	97.9	65.1	86.3	91.8	97.4	99.8	99.9	87.6	97.7	99.0

表 5. 与在 COCO 和 Flickr30K 数据集上微调的最先进图文检索方法的比较。$\text{BLIP}_{\text{CapFilt-L}}$ 使用由字幕生成器和过滤器借助 ViT-L 自举构建的数据集，对采用 ViT-B 骨干网络的模型进行预训练。

Method	Pre-train # Images	Flickr30K (1K test set) TR R@1	TR R@5	TR R@10	IR R@1	IR R@5	IR R@10
CLIP	400M	88.0	98.7	99.4	68.7	90.6	95.2
ALIGN	1.8B	88.6	98.7	99.7	75.7	93.8	96.8
ALBEF	14M	94.1	99.5	99.7	82.8	96.3	98.1
BLIP	14M	94.8	99.7	**100.0**	84.9	96.7	98.3
BLIP	129M	**96.0**	**99.9**	**100.0**	85.0	**96.8**	98.6
$\text{BLIP}_{\text{CapFilt-L}}$	129M	**96.0**	**99.9**	**100.0**	**85.5**	**96.8**	**98.7**
$\text{BLIP}_{\text{ViT-L}}$	129M	96.7	100.0	100.0	86.7	97.3	98.7

表 6. 在 Flickr30K 上的零样本图文检索结果。

共享除 SA 之外的层相比不共享能带来更好的性能，同时还能减小模型规模，从而提高训练效率。如果共享 SA 层，模型性能会因编码任务和解码任务之间的冲突而下降。

在 CapFilt 过程中，字幕生成器和过滤器分别在 COCO 上进行端到端微调。在表 4 中，我们研究了字幕生成器和过滤器以与预训练相同的方式共享参数时的影响。下游任务的性能有所下降，我们主要将其归因于 *confirmation bias*。由于参数共享，字幕生成器产生的噪声字幕不太可能被过滤器过滤掉，这一点可由较低的噪声比例（8%，相比之下为 25%）看出。

5. 与最先进方法的比较

在本节中，我们在广泛的视觉-语言下游任务[2]上将 BLIP 与现有的 VLP 方法进行比较。接下来我们简要介绍每项任务和微调策略。更多细节见附录。

5.1. 图像-文本检索

我们在 COCO 和 Flickr30K（Plummer 等，2015）数据集上评估 BLIP 的图像到文本检索（TR）和文本到图像检索（IR）性能。我们使用 ITC 和 ITM 损失对预训练模型进行微调。为实现更快的推理速度，我们遵循 Li 等（2021a）的方法，首先基于图像-文本特征相似度选择 k 个候选项，然后基于它们的成对 ITM 分数对所选候选项重新排序。我们将 COCO 的 $k=$ 设为 256，将 Flickr30K 的 $k=$ 设为 128。

如表 5 所示，与现有方法相比，BLIP 实现了显著的性能提升。在使用相同的 14M 预训练图像的情况下，BLIP 在 COCO 上的平均 recall@1 比此前最佳模型 ALBEF 高出 +2.7%。我们还通过将已在 COCO 上微调的模型直接迁移到 Flickr30K 来进行零样本检索。结果如表 6 所示，其中 BLIP 也以较大幅度优于现有方法。

5.2. 图像描述

我们考虑两个图像描述数据集：NoCaps（Agrawal 等，2019）和 COCO，二者均使用在 COCO 上通过 LM 损失微调的模型进行评估。与 Wang 等（2021）类似，我们在每个描述的开头添加提示语“a picture of”，这带来了略微更好的结果。如表 7 所示，使用 1400 万张预训练图像的 BLIP 显著优于使用相近数量预训练数据的方法。使用 1.29 亿张图像的 BLIP 达到了与 LEMON 相当的性能，

[2]we omit SNLI-VE from the benchmark because its test data has been reported to be noisy (Do et al., 2020)

Method	Pre-train #Images	NoCaps validation in-domain C	in-domain S	near-domain C	near-domain S	out-domain C	out-domain S	overall C	overall S	COCO Caption Karpathy test B@4	COCO Caption Karpathy test C
Enc-Dec (Changpinyo et al., 2021)	15M	92.6	12.5	88.3	12.1	94.5	11.9	90.2	12.1	-	110.9
VinVL† (Zhang et al., 2021)	5.7M	103.1	14.2	96.1	13.8	88.3	12.1	95.5	13.5	38.2	129.3
$\text{LEMON}_{\text{base}}$† (Hu et al., 2021)	12M	104.5	14.6	100.7	14.0	96.7	12.4	100.4	13.8	-	-
$\text{LEMON}_{\text{base}}$† (Hu et al., 2021)	200M	107.7	14.7	106.2	14.3	107.9	13.1	106.8	14.1	**40.3**	**133.3**
BLIP	14M	111.3	15.1	104.5	14.4	102.4	13.7	105.1	14.4	38.6	129.7
BLIP	129M	109.1	14.8	105.8	14.4	105.7	13.7	106.3	14.3	39.4	131.4
$\text{BLIP}_{\text{CapFilt-L}}$	129M	**111.8**	**14.9**	**108.6**	**14.8**	**111.5**	**14.2**	**109.6**	**14.7**	39.7	**133.3**
$\text{LEMON}_{\text{large}}$† (Hu et al., 2021)	200M	116.9	15.8	113.3	15.1	111.3	14.0	113.4	15.0	40.6	135.7
$\text{SimVLM}_{\text{huge}}$ (Wang et al., 2021)	1.8B	113.7	-	110.9	-	115.2	-	112.2	-	40.6	143.3
$\text{BLIP}_{\text{ViT-L}}$	129M	114.9	15.2	112.1	14.9	115.3	14.4	113.2	14.8	40.4	136.7

Table 7. Comparison with state-of-the-art image captioning methods on NoCaps and COCO Caption. All methods optimize the cross-entropy loss during finetuning. C: CIDEr, S: SPICE, B@4: BLEU@4. $\text{BLIP}_{\text{CapFilt-L}}$ is pre-trained on a dataset bootstrapped by captioner and filter with ViT-L. VinVL† and LEMON† require an object detector pre-trained on 2.5M images with human-annotated bounding boxes and high resolution (800×1333) input images. $\text{SimVLM}_{\text{huge}}$ uses 13× more training data and a larger vision backbone than ViT-L.

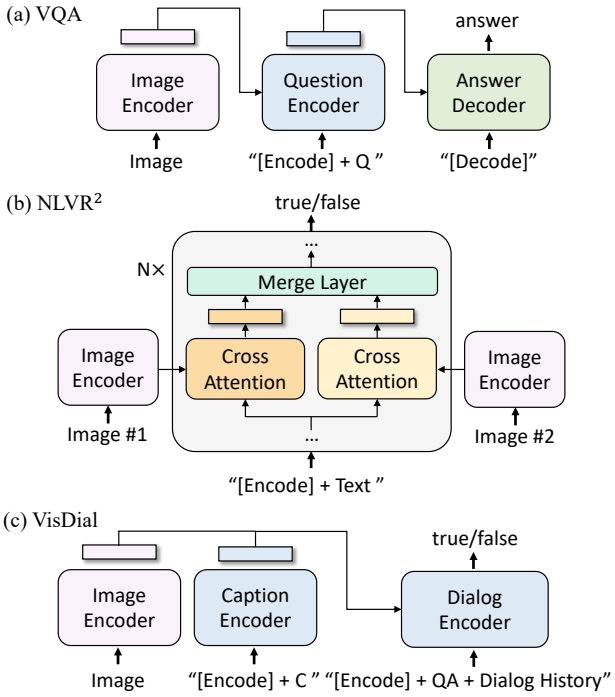


Figure 5. Model architecture for the downstream tasks. Q: question; C: caption; QA: question-answer pair.

200M images. Note that LEMON requires a computational-heavy pre-trained object detector and higher resolution (800×1333) input images, leading to substantially slower inference time than the detector-free BLIP which uses lower resolution (384×384) input images.

5.3. Visual Question Answering (VQA)

VQA (Antol et al., 2015) requires the model to predict an answer given an image and a question. Instead of formulating VQA as a multi-answer classification task (Chen et al., 2020; Li et al., 2020), we follow Li et al. (2021a) and consider it as an answer generation task, which enables open-ended VQA. As shown in Figure 5(a), during finetuning, we rearrange the pre-trained model, where an image-question is first encoded into multimodal embeddings and then given to an answer decoder. The VQA model is finetuned with the LM loss using ground-truth answers as targets.

Method	Pre-train #Images	VQA test-dev	VQA test-std	NLVR² dev	NLVR² test-P
LXMERT	180K	72.42	72.54	74.90	74.50
UNITER	4M	72.70	72.91	77.18	77.85
VL-T5/BART	180K	-	71.3	-	73.6
OSCAR	4M	73.16	73.44	78.07	78.36
SOHO	219K	73.25	73.47	76.37	77.32
VILLA	4M	73.59	73.67	78.39	79.30
UNIMO	5.6M	75.06	75.27	-	-
ALBEF	14M	75.84	76.04	82.55	83.14
$\text{SimVLM}_{\text{base}}$†	1.8B	77.87	78.14	81.72	81.77
BLIP	14M	77.54	77.62	**82.67**	82.30
BLIP	129M	78.24	78.17	82.48	**83.08**
$\text{BLIP}_{\text{CapFilt-L}}$	129M	**78.25**	**78.32**	82.15	82.24

Table 8. Comparison with state-of-the-art methods on VQA and NLVR². ALBEF performs an extra pre-training step for NLVR². SimVLM† uses 13× more training data and a larger vision backbone (ResNet+ViT) than BLIP.

The results are shown in Table 8. Using 14M images, BLIP outperforms ALBEF by +1.64% on the test set. Using 129M images, BLIP achieves better performance than SimVLM which uses 13× more pre-training data and a larger vision backbone with an additional convolution stage.

5.4. Natural Language Visual Reasoning (NLVR²)

NLVR² (Suhr et al., 2019) asks the model to predict whether a sentence describes a pair of images. In order to enable rea-

Method	Pre-train #Images	NoCaps validation in-domain C	in-domain S	near-domain C	near-domain S	out-domain C	out-domain S	overall C	overall S	COCO Caption Karpathy test B@4	COCO Caption Karpathy test C
Enc-Dec (Changpinyo et al., 2021)	15M	92.6	12.5	88.3	12.1	94.5	11.9	90.2	12.1	-	110.9
VinVL† (Zhang et al., 2021)	5.7M	103.1	14.2	96.1	13.8	88.3	12.1	95.5	13.5	38.2	129.3
$\text{LEMON}_{\text{base}}$† (Hu et al., 2021)	12M	104.5	14.6	100.7	14.0	96.7	12.4	100.4	13.8	-	-
$\text{LEMON}_{\text{base}}$† (Hu et al., 2021)	200M	107.7	14.7	106.2	14.3	107.9	13.1	106.8	14.1	**40.3**	**133.3**
BLIP	14M	111.3	15.1	104.5	14.4	102.4	13.7	105.1	14.4	38.6	129.7
BLIP	129M	109.1	14.8	105.8	14.4	105.7	13.7	106.3	14.3	39.4	131.4
$\text{BLIP}_{\text{CapFilt-L}}$	129M	**111.8**	**14.9**	**108.6**	**14.8**	**111.5**	**14.2**	**109.6**	**14.7**	39.7	**133.3**
$\text{LEMON}_{\text{large}}$† (Hu et al., 2021)	200M	116.9	15.8	113.3	15.1	111.3	14.0	113.4	15.0	40.6	135.7
$\text{SimVLM}_{\text{huge}}$ (Wang et al., 2021)	1.8B	113.7	-	110.9	-	115.2	-	112.2	-	40.6	143.3
$\text{BLIP}_{\text{ViT-L}}$	129M	114.9	15.2	112.1	14.9	115.3	14.4	113.2	14.8	40.4	136.7

表 7. 在 NoCaps 和 COCO Caption 上与最先进的图像描述方法的比较。所有方法在微调期间均优化交叉熵损失。C：CIDEr，S：SPICE，B@4：BLEU@4。$\text{BLIP}_{\text{CapFilt-L}}$ 在由描述生成器和过滤器引导构建的数据集上进行预训练，并使用 ViT-L。VinVL † 和 LEMON † 需要一个在 250 万张带有人类标注边界框的图像上预训练的目标检测器，以及高分辨率（800×1333）输入图像。$\text{SimVLM}_{\text{huge}}$ 使用比 ViT-L 多 13× 的训练数据和更大的视觉骨干网络。

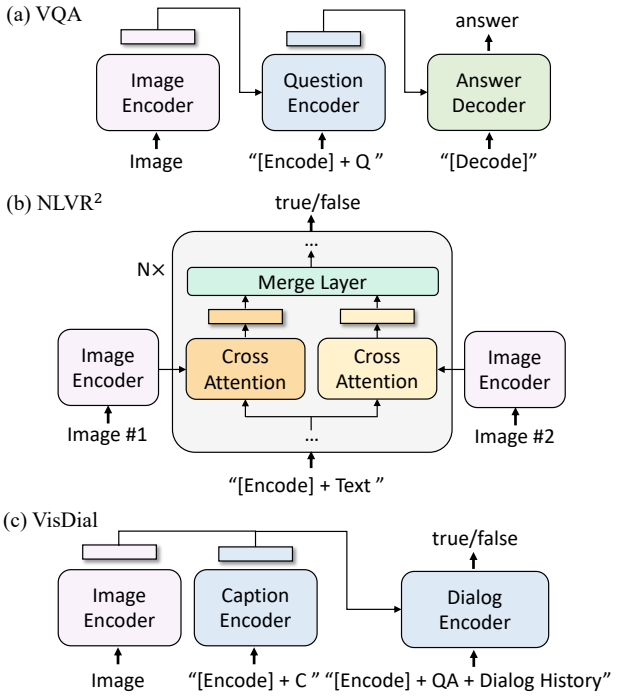


图 5. 下游任务的模型架构。Q：问题；C：图注；QA：问答对。

2亿张图像。需要注意的是，LEMON 需要计算量很大的预训练目标检测器以及更高分辨率（800×1333）的输入图像，因此其推理时间比无需检测器、使用较低分辨率（384×384）输入图像的 BLIP 慢得多。

5.3. 视觉问答（VQA）

VQA（Antol 等，2015）要求模型在给定一张图像和一个问题的情况下预测答案。与其将 VQA 表述为多答案分类任务（Chen 等，2020；

Method	Pre-train #Images	VQA test-dev	VQA test-std	NLVR² dev	NLVR² test-P
LXMERT	180K	72.42	72.54	74.90	74.50
UNITER	4M	72.70	72.91	77.18	77.85
VL-T5/BART	180K	-	71.3	-	73.6
OSCAR	4M	73.16	73.44	78.07	78.36
SOHO	219K	73.25	73.47	76.37	77.32
VILLA	4M	73.59	73.67	78.39	79.30
UNIMO	5.6M	75.06	75.27	-	-
ALBEF	14M	75.84	76.04	82.55	83.14
$\text{SimVLM}_{\text{base}}$†	1.8B	77.87	78.14	81.72	81.77
BLIP	14M	77.54	77.62	**82.67**	82.30
BLIP	129M	78.24	78.17	82.48	**83.08**
$\text{BLIP}_{\text{CapFilt-L}}$	129M	**78.25**	**78.32**	82.15	82.24

表 8. 在 VQA 和 NLVR^2 上与最先进方法的比较。ALBEF 针对 NLVR^2 进行了额外的预训练步骤。与 BLIP 相比，SimVLM † 使用了多 13× 的训练数据以及更大的视觉骨干网络（ResNet+ViT）。

Li 等人，2020），我们遵循 Li 等人（2021a）的做法，并将其视为一个答案生成任务，从而实现开放式 VQA。如图 5(a) 所示，在微调过程中，我们重新排列预训练模型，其中图像-问题首先被编码为多模态嵌入，然后输入到答案解码器。VQA 模型使用真实答案作为目标，并通过 LM 损失进行微调。

结果如表 8 所示。使用 14M 张图像时，BLIP 在测试集上比 ALBEF 高出 +1.64%。使用 129M 张图像时，BLIP 的性能优于 SimVLM，后者使用了多 13× 的预训练数据，并采用了更大的视觉骨干网络，且额外增加了一个卷积阶段。

5.4. 自然语言视觉推理（NLVR^2）

NLVR^2 (Suhr 等人，2019)要求模型预测一个句子是否描述了一对图像。为了实现推理-

Method	MRR↑	R@1↑	R@5↑	R@10↑	MR↓
VD-BERT	67.44	54.02	83.96	92.33	3.53
VD-ViLBERT†	69.10	55.88	85.50	93.29	3.25
BLIP	**69.41**	**56.44**	**85.90**	**93.30**	**3.20**

Table 9. Comparison with state-of-the-art methods on VisDial v1.0 validation set. VD-ViLBERT† (Murahari et al., 2020) pre-trains ViLBERT (Lu et al., 2019) with additional VQA data.

soning over two images, we make a simple modification to our pre-trained model which leads to a more computational-efficient architecture than previous approaches (Li et al., 2021a; Wang et al., 2021). As shown in Figure 5(b), for each transformer block in the image-grounded text encoder, there exist two cross-attention layers to process the two input images, and their outputs are merged and fed to the FFN. The two CA layers are intialized from the same pre-trained weights. The merge layer performs simple average pooling in the first 6 layers of the encoder, and performs concatenation followed by a linear projection in layer 6-12. An MLP classifier is applied on the output embedding of the `[Encode]` token. As shown in Table 8, BLIP outperforms all existing methods except for ALBEF which performs an extra step of customized pre-training. Interestingly, performance on NLVR^2 does not benefit much from additional web images, possibly due to the domain gap between web data and downstream data.

5.5. Visual Dialog (VisDial)

VisDial (Das et al., 2017) extends VQA in a natural conversational setting, where the model needs to predict an answer not only based on the image-question pair, but also considering the dialog history and the image's caption. We follow the discriminative setting where the model ranks a pool of answer candidates (Gan et al., 2019; Wang et al., 2020; Murahari et al., 2020). As shown in Figure 5(c), we concatenate image and caption embeddings, and pass them to the dialog encoder through cross-attention. The dialog encoder is trained with the ITM loss to discriminate whether the answer is true or false for a question, given the entire dialog history and the image-caption embeddings. As shown in Table 9, our method achieves state-of-the-art performance on VisDial v1.0 validation set.

5.6. Zero-shot Transfer to Video-Language Tasks

Our image-language model has strong generalization ability to video-language tasks. In Table 10 and Table 11, we perform zero-shot transfer to *text-to-video retrieval* and *video question answering*, where we directly evaluate the models trained on COCO-retrieval and VQA, respectively. To process video input, we uniformly sample n frames per video ($n = 8$ for retrieval and $n = 16$ for QA), and concatenate the frame features into a single sequence. Note that this simple approach ignores all temporal information.

Method	R1↑	R5↑	R10↑	MdR↓
zero-shot				
ActBERT (Zhu & Yang, 2020)	8.6	23.4	33.1	36
SupportSet (Patrick et al., 2021)	8.7	23.0	31.1	31
MIL-NCE (Miech et al., 2020)	9.9	24.0	32.4	29.5
VideoCLIP (Xu et al., 2021)	10.4	22.2	30.0	-
FiT (Bain et al., 2021)	18.7	39.5	51.6	10
BLIP	**43.3**	**65.6**	**74.7**	**2**
finetuning				
ClipBERT (Lei et al., 2021)	22.0	46.8	59.9	6
VideoCLIP (Xu et al., 2021)	30.9	55.4	66.8	-

Table 10. Comparisons with state-of-the-art methods for text-to-**video** retrieval on the 1k test split of the MSRVTT dataset.

Method	MSRVTT-QA	MSVD-QA
zero-shot		
VQA-T (Yang et al., 2021)	2.9	7.5
BLIP	19.2	35.2
finetuning		
HME (Fan et al., 2019)	33.0	33.7
HCRN (Le et al., 2020)	35.6	36.1
VQA-T (Yang et al., 2021)	41.5	46.3

Table 11. Comparisons with state-of-the-art methods for **video** question answering. We report top-1 test accuracy on two datasets.

Despite the domain difference and lack of temporal modeling, our models achieve state-of-the-art performance on both video-language tasks. For text-to-video retrieval, zero-shot BLIP even outperforms models finetuned on the target video dataset by +12.4% in recall@1. Further performance improvement can be achieved if the BLIP model is used to initialize a video-language model with temporal modeling (*e.g.* replace our ViT with a TimeSformer (Bertasius et al., 2021)) and finetuned on video data.

6. Additional Ablation Study

In this section, we provide additional ablation experiments on CapFilt.

Improvement with CapFilt is not due to longer training. Since the bootstrapped dataset contains more texts than the original dataset, training for the same number of epochs takes longer with the bootstrapped dataset. To verify that the effectiveness of CapFilt is not due to longer training, we replicate the web text in the original dataset so that it has the same number of training samples per epoch as the bootstrapped dataset. As shown in Table 12, longer training using the noisy web texts does not improve performance.

A new model should be trained on the bootstrapped dataset. The bootstrapped dataset is used to pre-train a new model. We investigate the effect of continue training

Method	MRR↑	R@1↑	R@5↑	R@10↑	MR↓
VD-BERT	67.44	54.02	83.96	92.33	3.53
VD-ViLBERT†	69.10	55.88	85.50	93.29	3.25
BLIP	**69.41**	**56.44**	**85.90**	**93.30**	**3.20**

表 9. 在 VisDial v1.0 验证集上与最先进方法的比较。VD-ViLBERT †（Murahari 等，2020）使用额外的 VQA 数据对 ViLBERT（Lu 等，2019）进行预训练。

在两张图像上进行推理时，我们对预训练模型进行了简单修改，从而得到一种比以往方法（Li et al., 2021a；Wang et al., 2021）计算效率更高的架构。如图 5(b) 所示，对于图像引导文本编码器中的每个 transformer 块，都有两个交叉注意力层用于处理两张输入图像，它们的输出会被合并并输入到 FFN。这两个 CA 层使用相同的预训练权重进行初始化。合并层在编码器的前 6 层中执行简单的平均池化，在第 6–12 层中执行拼接后接线性投影。一个 MLP 分类器应用于 [Encode] token 的输出嵌入。如表 8 所示，BLIP 优于除 ALBEF 之外的所有现有方法，而 ALBEF 额外进行了定制化预训练步骤。有趣的是，NLVR^2 上的性能并未从额外的网络图像中获得太多提升，这可能是由于网络数据与下游数据之间存在领域差距。

5.5. 视觉对话（VisDial）

VisDial（Das 等，2017）将 VQA 扩展到自然对话场景中，在该场景下，模型不仅需要基于图像-问题对来预测答案，还需要考虑对话历史和图像的标题。我们遵循判别式设置，其中模型对一组候选答案进行排序（Gan 等，2019；Wang 等，2020；Murahari 等，2020）。如图 5(c) 所示，我们将图像和标题嵌入进行拼接，并通过交叉注意力将它们传递给对话编码器。在给定完整对话历史以及图像-标题嵌入的情况下，对话编码器使用 ITM 损失进行训练，以判别某个答案对于问题而言是真还是假。如表 9 所示，我们的方法在 VisDial v1.0 验证集上实现了最先进的性能。

5.6. 零样本迁移到视频-语言任务

我们的图像-语言模型对视频-语言任务具有很强的泛化能力。在表 10 和表 11 中，我们对 *text-to-video retrieval* 和 *video question answering* 进行零样本迁移，在其中分别直接评估在 COCO 检索和 VQA 上训练的模型。为处理视频输入，我们从每个视频中均匀采样 n 帧（检索任务中 $n = 8$，QA 任务中 $n = 16$），并将帧特征拼接成一个单一序列。请注意，这种简单方法忽略了所有时间信息。

Method	R1↑	R5↑	R10↑	MdR↓
zero-shot				
ActBERT (Zhu & Yang, 2020)	8.6	23.4	33.1	36
SupportSet (Patrick et al., 2021)	8.7	23.0	31.1	31
MIL-NCE (Miech et al., 2020)	9.9	24.0	32.4	29.5
VideoCLIP (Xu et al., 2021)	10.4	22.2	30.0	-
FiT (Bain et al., 2021)	18.7	39.5	51.6	10
BLIP	**43.3**	**65.6**	**74.7**	**2**
finetuning				
ClipBERT (Lei et al., 2021)	22.0	46.8	59.9	6
VideoCLIP (Xu et al., 2021)	30.9	55.4	66.8	-

表 10. 在 MSRVTT 数据集的 1k 测试划分上，与用于文本到视频检索的最先进方法的比较。

Method	MSRVTT-QA	MSVD-QA
zero-shot		
VQA-T (Yang et al., 2021)	2.9	7.5
BLIP	19.2	35.2
finetuning		
HME (Fan et al., 2019)	33.0	33.7
HCRN (Le et al., 2020)	35.6	36.1
VQA-T (Yang et al., 2021)	41.5	46.3

表 11. 与视频问答任务最先进方法的比较。我们报告了在两个数据集上的 top-1 测试准确率。

尽管存在领域差异且缺乏时序建模，我们的模型在两个视频-语言任务上都达到了最先进的性能。对于文本到视频检索，零样本 BLIP 甚至在 recall@1 上比在目标视频数据集上微调的模型高出 +12.4%。如果使用 BLIP 模型来初始化一个具有时序建模能力的视频-语言模型（*e.g.* 将我们的 ViT 替换为 TimeSformer（Bertasius et al., 2021）），并在视频数据上进行微调，则可以进一步提升性能。

6. 补充消融实验

在本节中，我们提供了关于 CapFilt 的额外消融实验。

CapFilt 带来的提升并非源于更长的训练时间。由于自举数据集包含的文本多于原始数据集，在相同 epoch 数下使用自举数据集进行训练会花费更长时间。为验证 CapFilt 的有效性并非源于更长的训练时间，我们复制原始数据集中的网络文本，使其每个 epoch 的训练样本数量与自举数据集相同。如表 12 所示，使用有噪声的网络文本进行更长时间的训练并不能提升性能。

应在自助采样数据集上训练一个新模型。该自助采样数据集用于预训练一个新模型。我们研究继续训练的效果。

CapFilt	#Texts	Retrieval-FT (COCO)		Retrieval-ZS (Flickr)		Caption-FT (COCO)		Caption-ZS (NoCaps)	
		TR@1	IR@1	TR@1	IR@1	B@4	CIDEr	CIDEr	SPICE
No	15.3M	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
No	24.7M	78.3	60.5	93.7	82.2	37.9	127.7	102.1	14.0
Yes	24.7M	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

Table 12. The original web texts are replicated to have the same number of samples per epoch as the bootstrapped dataset. Results verify that the improvement from CapFilt is not due to longer training time.

Continue	Retrieval-FT (COCO)		Retrieval-ZS (Flickr)		Caption-FT (COCO)		Caption-ZS (NoCaps)	
	TR@1	IR@1	TR@1	IR@1	B@4	CIDEr	CIDEr	SPICE
Yes	80.6	63.0	94.5	84.6	38.5	129.9	104.5	14.2
No	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

Table 13. Continue training the pre-trained model offers less gain compared to training a new model with the bootstrapped dataset.

from the previous pre-trained model, using the bootstrapped dataset. Table 13 hows that continue training does not help. This observation agrees with the common practice in knowledge distillation, where the student model cannot be initialized from the teacher.

7. Conclusion

We propose BLIP, a new VLP framework with state-of-the-art performance on a wide range of downstream vision-language tasks, including understanding-based and generation-based tasks. BLIP pre-trains a multimodal mixture of encoder-decoder model using a dataset bootstrapped from large-scale noisy image-text pairs by injecting diverse synthetic captions and removing noisy captions. Our bootstrapped dataset are released to facilitate future vision-language research.

There are a few potential directions that can further enhance the performance of BLIP: (1) Multiple rounds of dataset bootstrapping; (2) Generate multiple synthetic captions per image to further enlarge the pre-training corpus; (3) Model ensemble by training multiple different captioners and filters and combining their forces in CapFilt. We hope that our paper motivates future work to focus on making improvements in both the model aspect and the data aspect, the bread and butter of vision-language research.

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CapFilt	#Texts	Retrieval-FT (COCO)		Retrieval-ZS (Flickr)		Caption-FT (COCO)		Caption-ZS (NoCaps)	
		TR@1	IR@1	TR@1	IR@1	B@4	CIDEr	CIDEr	SPICE
No	15.3M	78.4	60.7	93.9	82.1	38.0	127.8	102.2	13.9
No	24.7M	78.3	60.5	93.7	82.2	37.9	127.7	102.1	14.0
Yes	24.7M	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

表 12。原始网页文本被重复，以使每个 epoch 的样本数量与自助采样数据集相同。结果验证了 CapFilt 带来的提升并非源于更长的训练时间。

Continue	Retrieval-FT (COCO)		Retrieval-ZS (Flickr)		Caption-FT (COCO)		Caption-ZS (NoCaps)	
	TR@1	IR@1	TR@1	IR@1	B@4	CIDEr	CIDEr	SPICE
Yes	80.6	63.0	94.5	84.6	38.5	129.9	104.5	14.2
No	80.6	63.1	94.8	84.9	38.6	129.7	105.1	14.4

表13。与使用自举数据集训练新模型相比，继续训练预训练模型带来的增益较小。

使用自举数据集，从先前的预训练模型开始训练。表 13 显示，继续训练并没有帮助。这一观察结果与知识蒸馏中的常见做法一致，即学生模型不能从教师模型初始化。

7. 结论

我们提出了 BLIP，这是一种新的 VLP 框架，在广泛的下游视觉-语言任务上实现了最先进的性能，包括基于理解的任务和基于生成的任务。BLIP 使用一个从大规模噪声图像-文本对中通过注入多样化的合成描述并移除噪声描述而自举得到的数据集，对多模态编码器-解码器混合模型进行预训练。我们发布了自举数据集，以促进未来的视觉-语言研究。

有几个潜在方向可以进一步提升 BLIP 的性能：（1）进行多轮数据集自举；（2）为每张图像生成多个合成描述，以进一步扩大预训练语料库；（3）通过训练多个不同的描述生成器和过滤器，并在 CapFilt 中结合它们的能力来进行模型集成。我们希望本文能够激励未来工作同时关注模型层面和数据层面的改进，这是视觉-语言研究的核心所在。

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A. Downstream Task Details

Table 14 shows the hyperparameters that we use for finetuning on the downstream vision-language tasks. All tasks uses AdamW optimizer with a weight decay of 0.05 and a cosine learning rate schedule. We use an image resolution of 384×384, except for VQA where we follow Wang et al. (2021) and use 480×480 images. Next we delineate the dataset details.

Image-Text Retrieval. We use the Karpathy split (Karpathy & Li, 2015) for both COCO and Flickr30K. COCO contains 113/5k/5k images for train/validation/test, and Flickr30K contains 29k/1k/1k images for train/validation/test.

Image Captioning. We finetune on COCO's Karpathy train split, and evaluate on COCO's Karpathy test split and NoCaps validation split. During inference, we use beam search with a beam size of 3, and set the maximum generation length as 20.

VQA. We experiment with the VQA2.0 dataset (Goyal et al., 2017), which contains 83k/41k/81k images for training/validation/test. Following Li et al. (2021a), we use both training and validation splits for training, and include additional training samples from Visual Genome. During inference on VQA, we use the decoder to rank the 3,128 candidate answers (Li et al., 2021a; Kim et al., 2018).

NLVR^2. We conduct experiment on the official split (Suhr et al., 2019).

VisDial. We finetune on the training split of VisDial v1.0 and evaluate on its validation set.

Task	init LR (ViT-L)	batch size	#epoch
Retrieval	$1e^{-5}$ ($5e^{-6}$)	256	6
Captioning	$1e^{-5}$ ($2e^{-6}$)	256	5
VQA	$2e^{-5}$	256	10
NLVR^2	$3e^{-5}$	256	15
VisDial	$2e^{-5}$	240	20

Table 14. Finetuning hyperparameters for downstream tasks.

B. Additional Examples of Synthetic Captions

In Figure 6, we show additional examples of images and texts where the web captions are filtered out, and the synthetic captions are kept as clean training samples.

C. Pre-training Dataset Details

Table 15 shows the statistics of the pre-training datasets.

	COCO	VG	SBU	CC3M	CC12M	LAION
# image	113K	100K	860K	3M	10M	115M
# text	567K	769K	860K	3M	10M	115M

Table 15. Statistics of the pre-training datasets.



Figure 6. Examples of the web text T_w and the synthetic text T_s. Green texts are accepted by the filter, whereas red texts are rejected.

A. 下游任务详情

表14展示了我们用于在下游视觉-语言任务上进行微调的超参数。所有任务均使用 AdamW 优化器，权重衰减为 0.05，并采用余弦学习率调度。我们使用 384 × 384 的图像分辨率，但 VQA 除外，我们遵循 Wang 等人（2021）的做法，使用 480 × 480 的图像。接下来我们详细说明数据集细节。

图文检索。我们对 COCO 和 Flickr30K 均使用 Karpathy 划分（Karpathy & Li，2015）。COCO 包含 113/5k/5k 张图像，分别用于训练/验证/测试；Flickr30K 包含 29k/1k/1k 张图像，分别用于训练/验证/测试。

图像字幕生成。我们在 COCO 的 Karpathy 训练划分上进行微调，并在 COCO 的 Karpathy 测试划分和 NoCaps 验证划分上进行评估。在推理过程中，我们使用束大小为 3 的束搜索，并将最大生成长度设为 20。

VQA。我们在 VQA2.0 数据集（Goyal 等，2017）上进行实验，该数据集包含 83k/41k/81k 张图像，分别用于训练/验证/测试。遵循 Li 等（2021a），我们同时使用训练集和验证集进行训练，并包含来自 Visual Genome 的额外训练样本。在 VQA 推理过程中，我们使用解码器对 3,128 个候选答案进行排序（Li 等，2021a；Kim 等，2018）。

NLVR^2。我们在官方划分（Suhr 等，2019）上进行实验。

VisDial。我们在 VisDial v1.0 的训练集上进行微调，并在其验证集上进行评估。

Task	init LR (ViT-L)	batch size	#epoch
Retrieval	$1e^{-5}$ ($5e^{-6}$)	256	6
Captioning	$1e^{-5}$ ($2e^{-6}$)	256	5
VQA	$2e^{-5}$	256	10
NLVR^2	$3e^{-5}$	256	15
VisDial	$2e^{-5}$	240	20

表 14. 下游任务的微调超参数。

B. 合成字幕的更多示例

在图 6 中，我们展示了更多图像和文本示例，其中网页标题被过滤掉，而合成标题被保留为干净的训练样本。

C. 预训练数据集详情

表 15 显示了预训练数据集的统计信息。

	COCO	VG	SBU	CC3M	CC12M	LAION
# image	113K	100K	860K	3M	10M	115M
# text	567K	769K	860K	3M	10M	115M

表 15. 预训练数据集的统计信息。

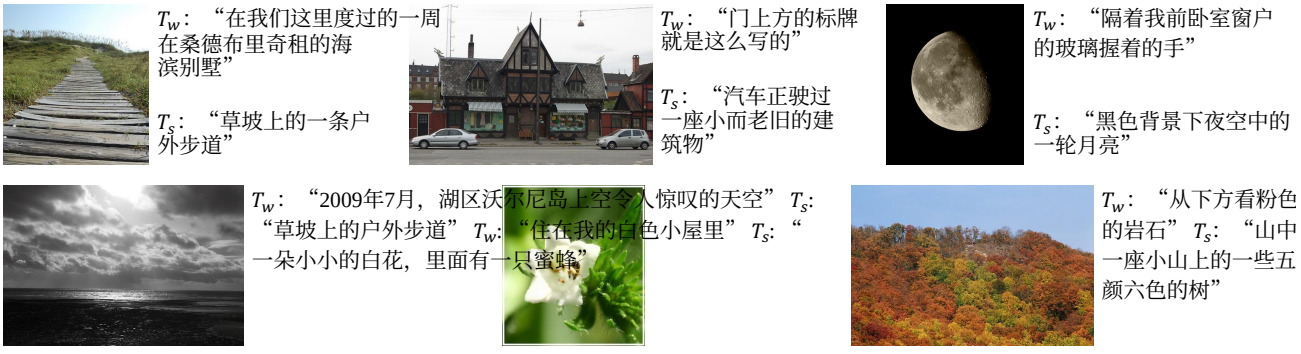


图 6. 网络文本 T_w 和合成文本 T_s 的示例。绿色文本被过滤器接受，而红色文本被拒绝。